

Class 2 – Ch 4- Computer Programming

- Conditional Statement – If-Else ✓
- Switch-Case ✓
- Loops – Do, While, Do-While
- Break & Continue
- ✓ Parameter Parsing

if (else)
int x = 12345
if (x > 0)
print (positive);

if ($x > 0$)

positive

else

Negative

if ($x > 0$)

positive

else if ($x > 5$)

Greater than 5

else if ($x > 100$)

else —

1 input $x = 4$

which one is answer

✓
Case 1 Option 1
Case 2 Option ②
Case 3 Option boom

default — ☐

```

int day = 2;
printf("I.D; & day");

switch(day) {
    case 1:
        printf("Saturday");
        break;

    case 2:
        printf("Sunday");
        break;

    case 3:
        printf("Monday");
        break;

    case 4:
        printf("Tuesday");
        break;

    default:
        printf("Invalid Day");
}
    
```

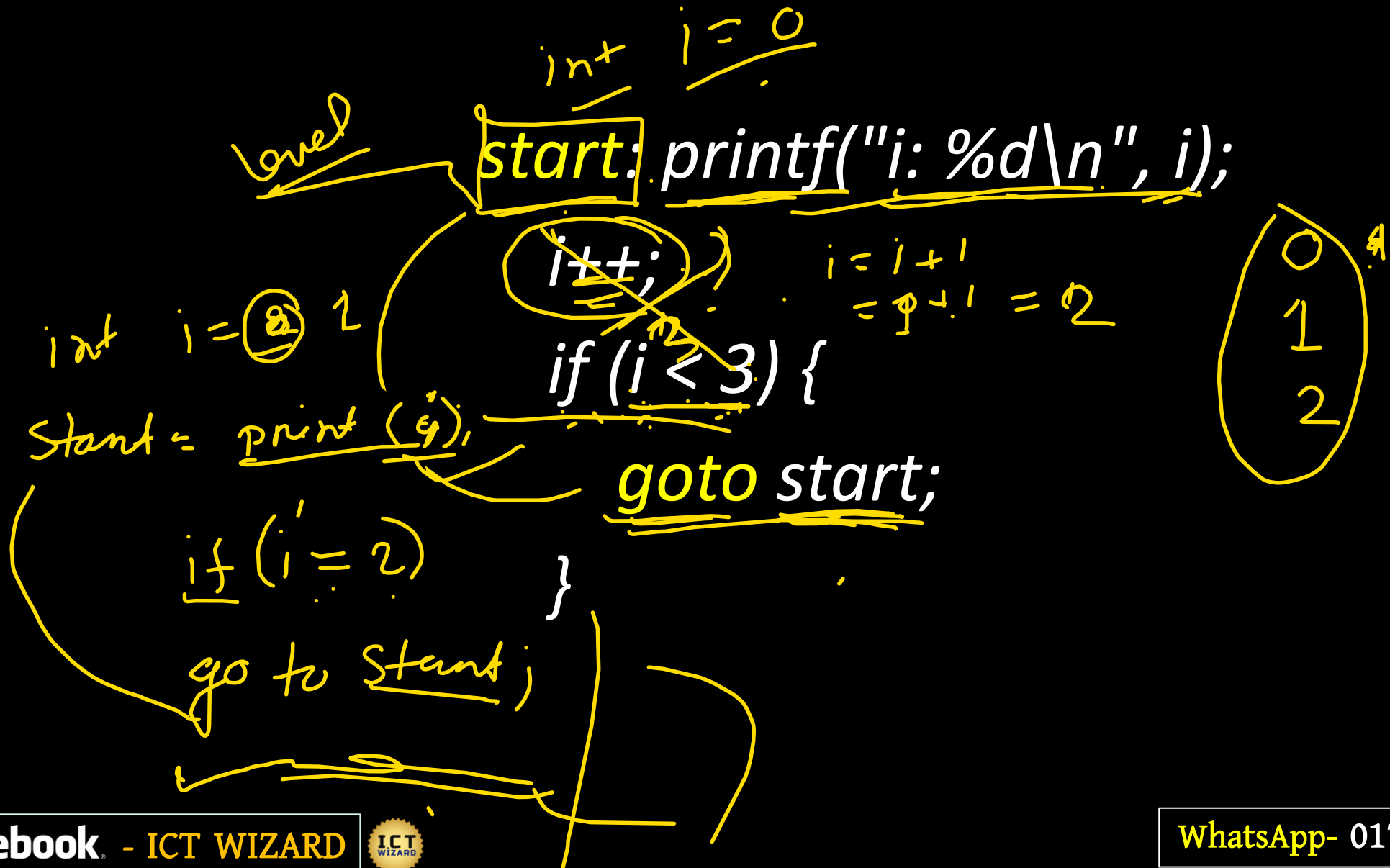
Handwritten annotations:

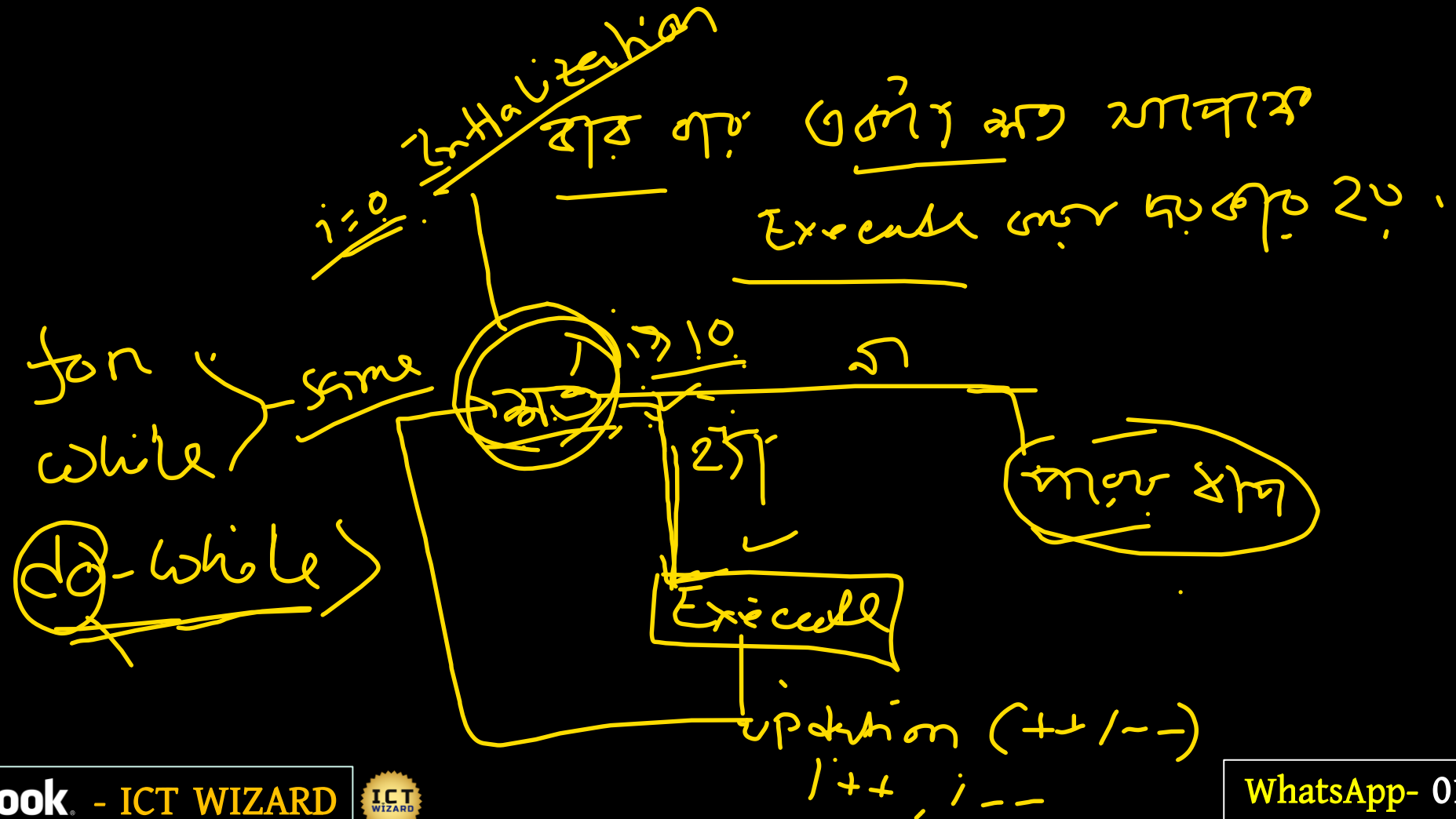
- A yellow circle around the variable `day` in the declaration `int day = 2;`.
- A yellow circle around the value `2` in the `switch(day)` statement.
- A yellow arrow pointing from the circled `2` in the `switch` statement to the `case 2:` block.
- Handwritten text `Sunday` with an arrow pointing to the `printf("Sunday");` line.
- Handwritten text `case 1:` with a checkmark next to it.
- Handwritten text `case 3:` with a checkmark next to it.
- Handwritten text `case 4:` with a checkmark next to it.
- Handwritten text `default:` with a checkmark next to it.
- Handwritten text `printf("I.D; & day");` with a checkmark next to it.
- Handwritten text `break;` with a checkmark next to it.
- Handwritten text `printf("Saturday");` with a checkmark next to it.
- Handwritten text `printf("Monday");` with a checkmark next to it.
- Handwritten text `printf("Tuesday");` with a checkmark next to it.
- Handwritten text `printf("Invalid Day");` with a checkmark next to it.

<u>if-else</u>	<u>switch</u>
<u>Condition</u> পরীক্ষা করে $x > 0$	নির্দিষ্ট <u>value</u> পরীক্ষা করে
<u>>, <, >=, <=</u> ব্যবহার করা যায়	সাধারণত নির্দিষ্ট <u>value-এর</u> সাথে <u>match</u> করে
<u>Complex condition</u> করা যায়	<u>Multiple choice-এর</u> জন্য সুবিধাজনক
<u>Range check</u> করা যায় if $x > 0$ এ $x < 10$	সরাসরি <u>range check</u> করা যায় না ১৬
<u>break</u> প্রয়োজন নেই	সাধারণত <u>break</u> ব্যবহার করা হয়

> Jump statement - Goto

level: sum }
it ~~~~~ }
goto level name





for(i = 0; i < 3; i++) {
 printf("IOT Wizard");
}

$i = i + 1$
 $= 0 + 1 = 1 + 1 = 2 + 1 = 3$

IOT Wizard
"
"

> While Loop

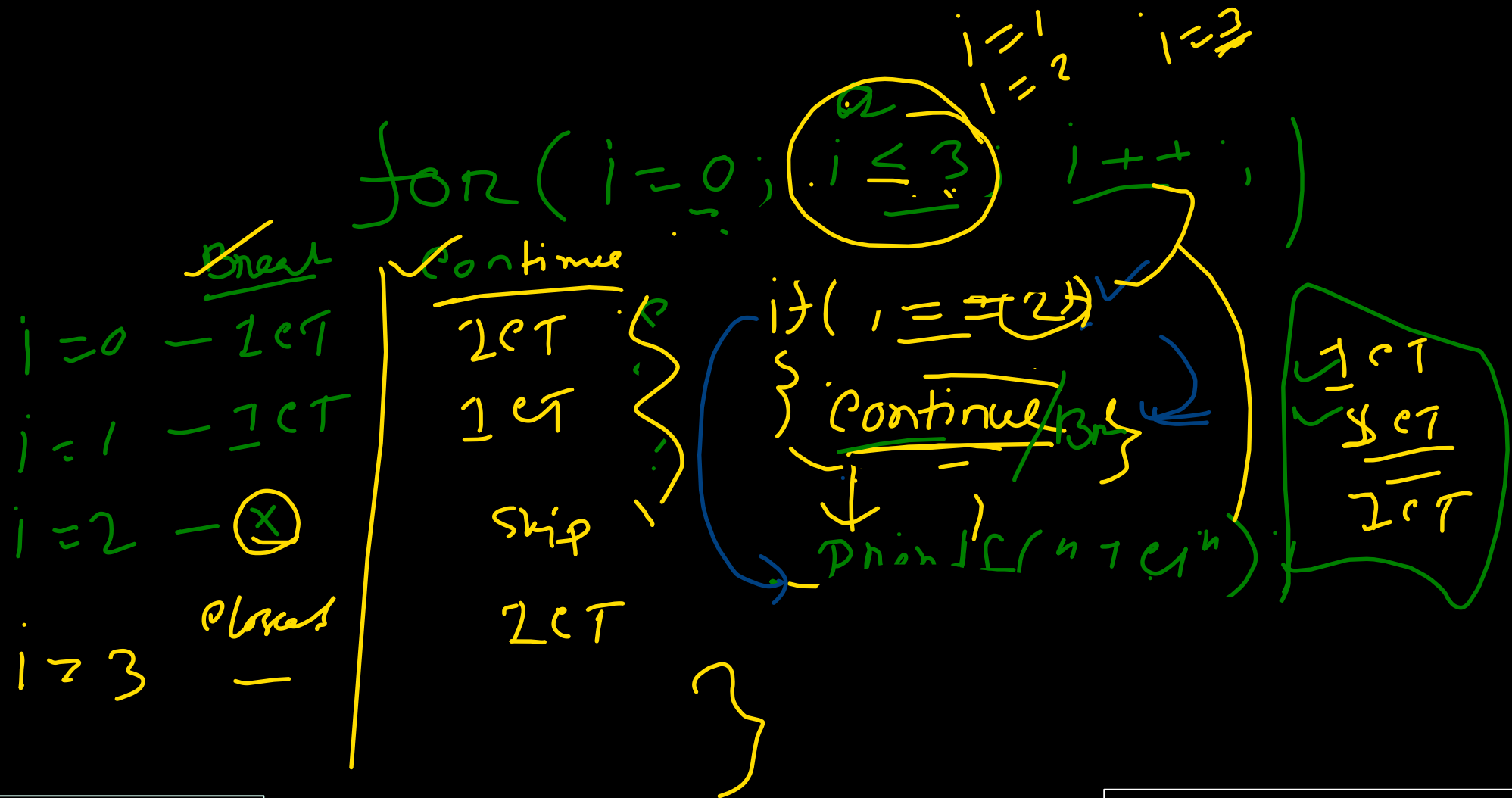
```
i = 0; 5
while(i < 3) {
    printf("ZCT Wizard");
    i++;
}
```

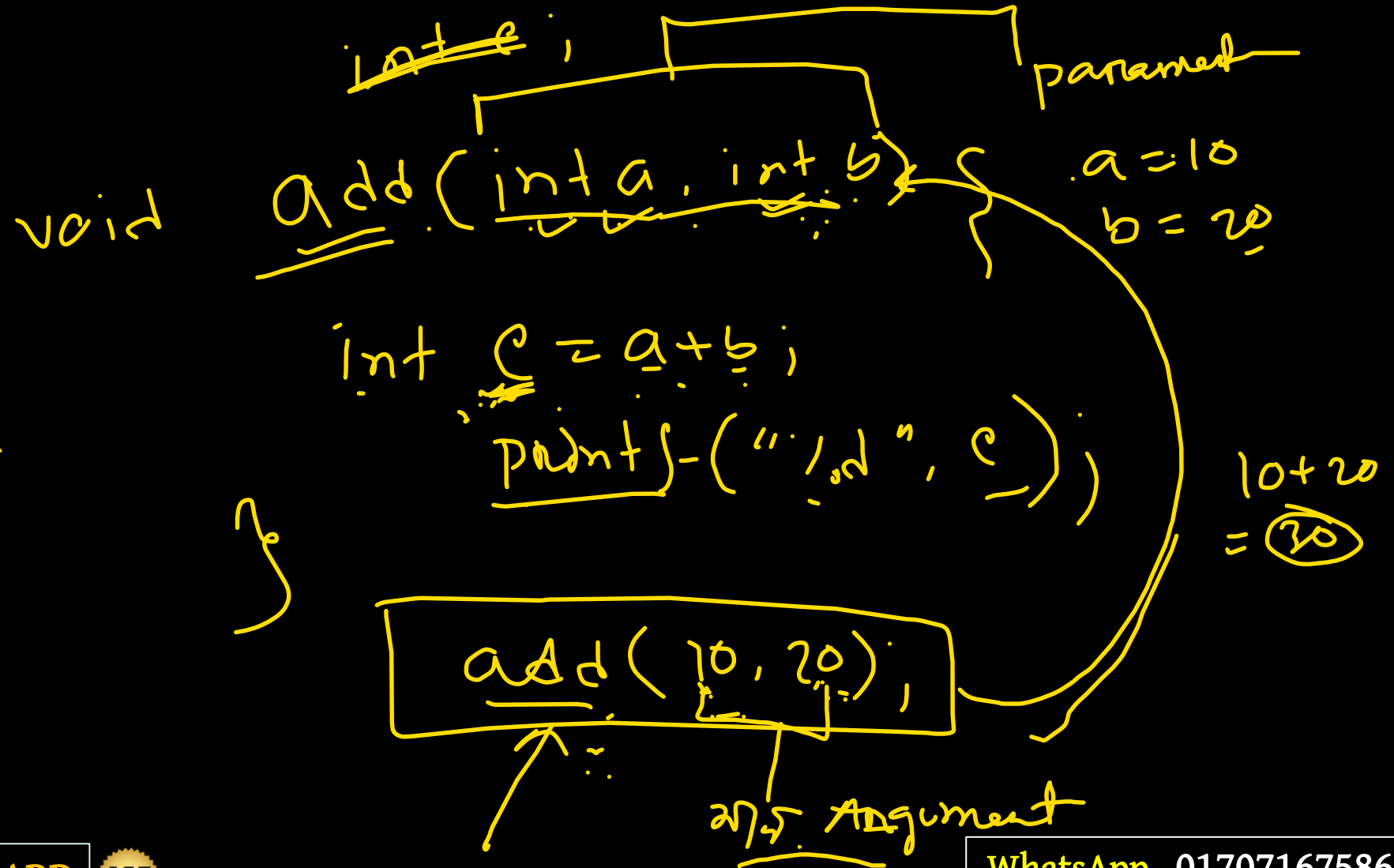
Handwritten code snippet illustrating a While Loop. The loop condition is `i < 3`, and the loop body contains `printf("ZCT Wizard");` and `i++`. The variable `i` is initialized to 0. The loop will execute as long as `i` is less than 3.

> Do-While Loop

```
i = 5;
do {
    printf("201 w\n");
    i++;
} while (i < 3);
```

বিষয়	for	while	do-while
Condition কখন পরীক্ষা হয়?	আগে	আগে	পরে
কমপক্ষে ১ বার চলবে?	✗ — ✓	✗ — ✓	✓ —
Initialization সাধারণত কোথায়?	<u>for-এর মধ্যে</u>	আগে	আগে
Update সাধারণত কোথায়?	for-এর মধ্যে	body-তে	body-তে
নির্দিষ্ট সংখ্যক বার চালাতে	খুব উপযোগী	উপযোগী	উপযোগী





> Parameter Parsing > Call by Value

```
void add(int x, int y)
{
```

```
  x = x + 10;
```

```
  y = y + 10;
```

```
}
```

```
  printf(" %d %d", x, y);
```

```
int main()
```

```
{
```

```
  int a = 10, b = 20;
```

```
  add(a, b);
```

```
}
```

output = 20, 30

10, 20

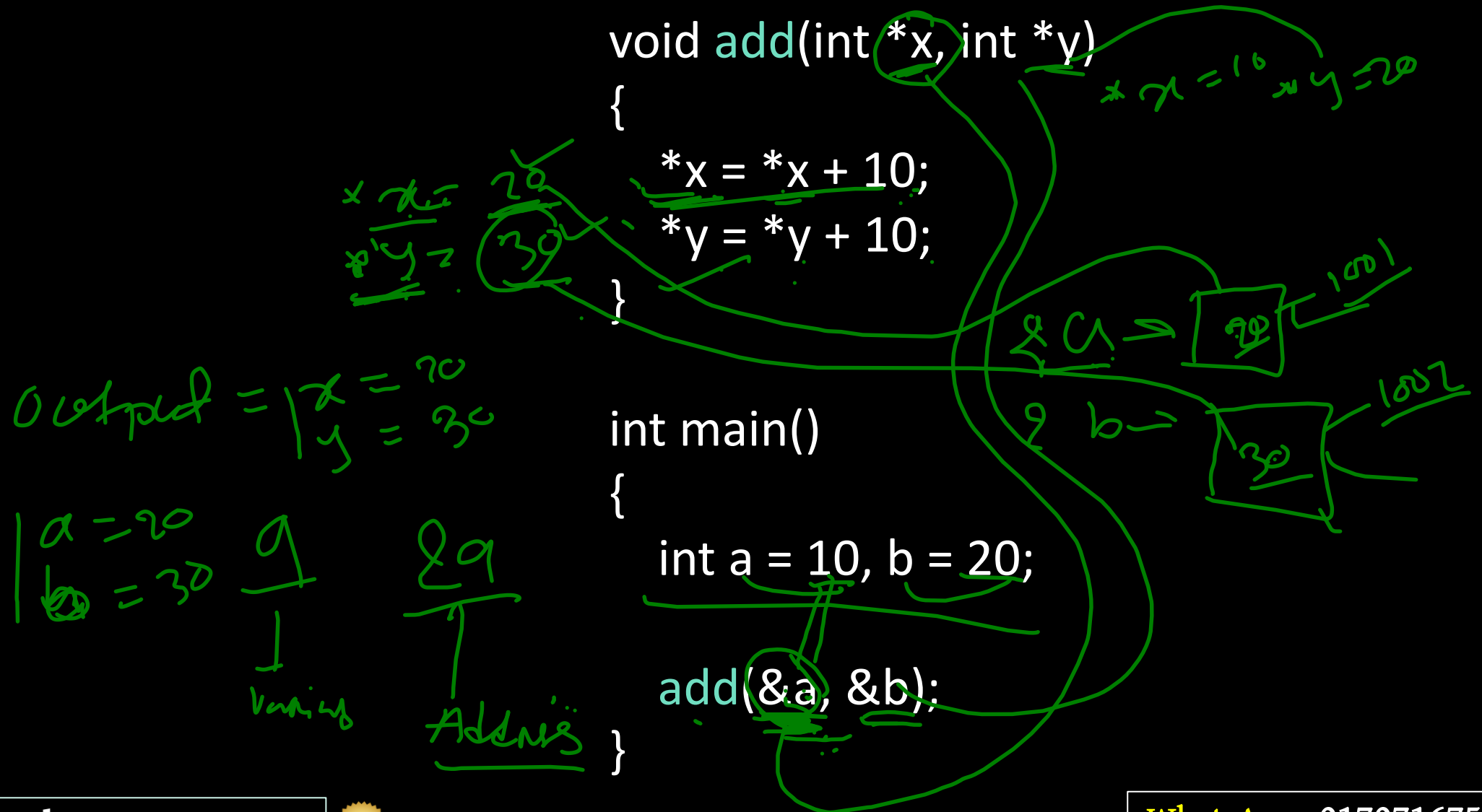
$x = 10$
 $y = 20$
 $x + 10 = 20$
 $y + 10 = 30$

20, 30

20

10

printf(" %d %d", x, y);



What will be the output of the following C code? **[COMBINED BANK-SO-IT-24]**

```
int main() {  
    int k = 0;  
    for (k = 0; k < 3; k++)  
        printf("multi");  
    return 0;  
}
```

Handwritten notes: *2nd i=0*, *1*, *2*

- କ) run time error ଧ) multi is printed infinitely
ଗ) multi is printed twice ଘ) multi is printed thrice

What will be the output of the following C code? [**Executive Trainee (Software), NPPCBL**]

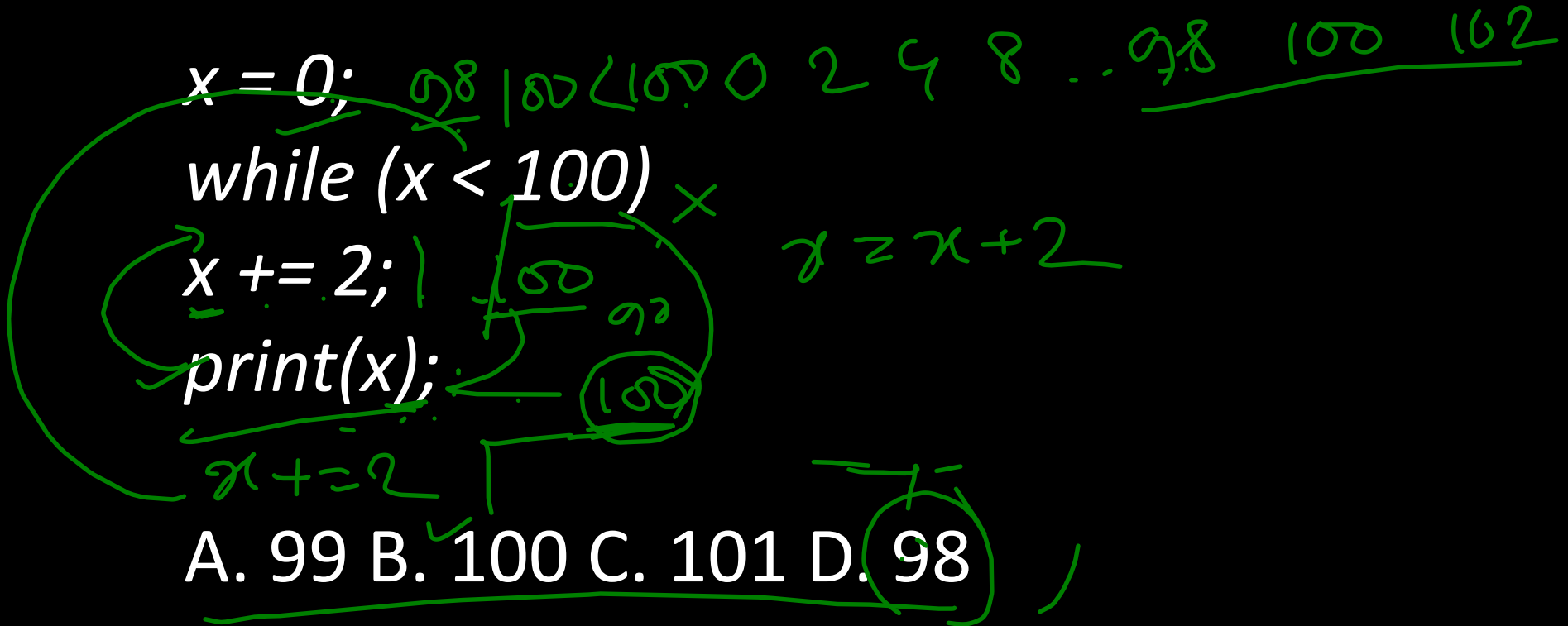
`x = 0;`

`while (x < 100)`

`x += 2;`

`print(x);`

A. 99 B. 100 C. 101 D. 98



What will be the output of the following C code? [AP, COB-23]

```
int i=i++, j=j++, k=k++;  
printf("%d %d %d", i, j, k);
```

A. 111 B. 000 C. garbage values D. Error

int i; int i;

What will be the output of the following C code? [SO IT CoB-24]

```
for(i=0; i<5; i++)  
    for(j=0; j<i; j++)  
        count++;
```

Handwritten notes and diagrams:

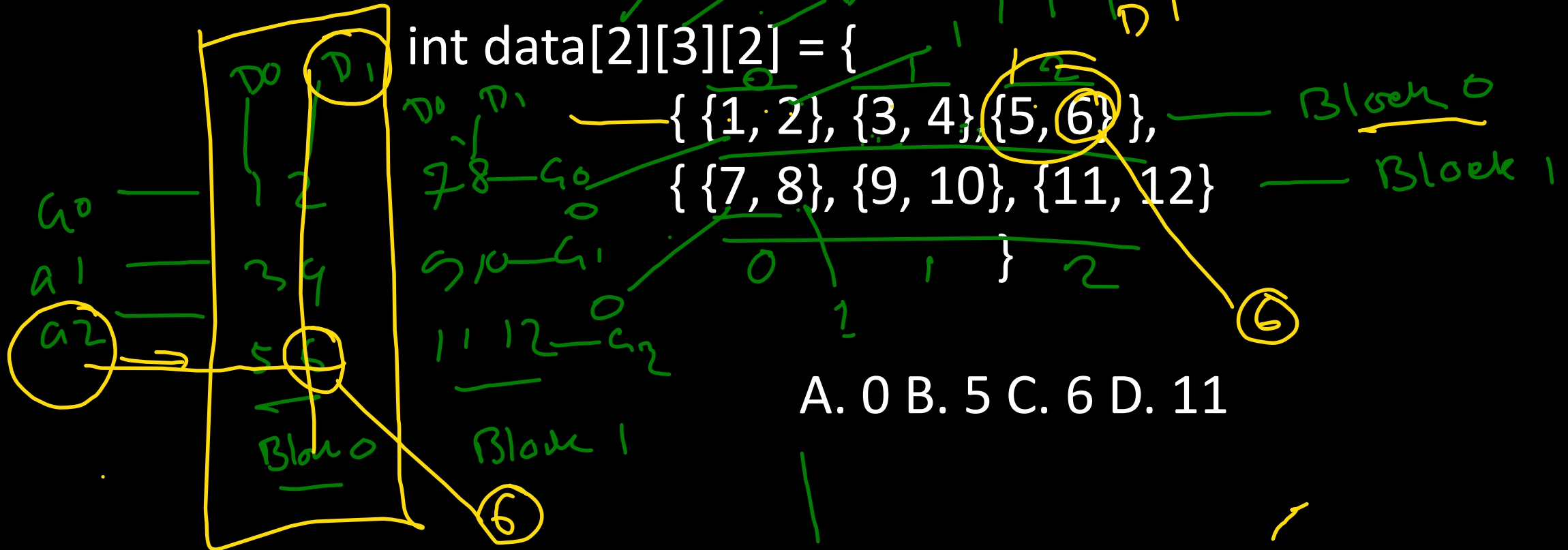
- $2 < 0$ (circled)
- Diagram showing nested loops for i and j with values 0, 1, 2, 3, 4.
- $count++$ (circled)
- $count = count + 1$ (boxed)
- $0 + 21$
- $0 + 1 = 1$
- $1 < 0 < 1$
- $0 < 5$
- 6 ± 10 (circled)

A. 1 B. 5 C. 10 D. 25

What will be the output of data[0][2][1]? - **SO IT CoB-24**

```
int data[2][3][2] = {  
    { {1, 2}, {3, 4}, {5, 6} },  
    { {7, 8}, {9, 10}, {11, 12} }
```

A. 0 B. 5 C. 6 D. 11



What is a loop? Describe its structure with a diagram. Marks – 5 (16th NTRCA)

1. What is the difference between Call by Value and Call by Reference? Marks – 3 (15th NTRCA)

2. What is Parameter Passing? Explain with an example. Marks – 2 (15th NTRCA)

3. What is the difference between ++a and a++? Explain with an example. Marks – 3 (15th NTRCA)